Universität Zürich[UZH]

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

DaST
Data • (Systems+Theory)

VLG Computer Vision and Learning Group

Bachelor's Thesis

Submission of the thesis
June 8, 2026

Advancing (Sparse) Multi-View Depth Estimation

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Abstract

Zusammenfassung

Acknowledgements

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Contents

1 Introduction

Recovering the three-dimensional structure of a dynamic scene from visual observations is a foundational challenge in computer vision with direct implications for robotics and autonomous navigation [5]. Recent spatial foundation models—such as MASt3R-SfM [7], VGGT [13], PI3, and PI3X [15]—have achieved remarkable per-frame geometric fidelity by regressing dense 3D pointmaps directly from images. Yet in 4D settings, where temporal coherence across a video sequence is as critical as spatial accuracy, these models fail: assembled frame-by-frame, their outputs exhibit severe high-frequency jitter that renders them unsuitable for robotics applications requiring smooth, metric-consistent depth. Geometry-refinement approaches such as GGPT [2] attempt to correct these inconsistencies by grounding spatial predictions with sparse SfM guidance, but fail to generalise to sparse-view dynamic scenes, worsening Chamfer distance by 3–10×. Conversely, dedicated 4D architectures—VGGT4D [6] and MonoFusion [16]—sacrifice geometric precision to buy temporal smoothness, inflating Chamfer distance by 20–160 % and producing systematic alignment failures in sparse-view settings.

The central finding of this thesis is that a *decoupled* paradigm outperforms both families: take a strong per-frame 3D estimator, align its outputs into a common coordinate frame through a lightweight temporal alignment strategy, and apply a confidence-weighted, mask-aware Gaussian post-processing filter (Opt) to enforce static-scene consistency. This approach reduces jitter by 25–81 % across four models (MASt3R-SfM, VGGT, PI3, PI3X) while preserving geometry to within ±2–8 % Chamfer distance—a strictly superior trade-off to every native 4D architecture and geometry-refinement baseline we evaluate. At the two-camera extreme, VGGT combined with Opt beats the native 4D architecture VGGT4D on every single metric.

1.1 Background and Motivation

Classical multi-view reconstruction pipelines such as Structure-from-Motion (SfM) [10] produce accurate, metric-scale geometry in well-constrained settings: they rely on repeatable keypoint matching and bundle adjustment, which deliver precise camera poses and sparse point clouds when images share sufficient texture, overlap, and baseline. In sparse-view configurations of two to four cameras, however, these conditions frequently fail. Low inter-camera overlap means that few keypoints are visible in more than one view simultaneously [5]; short or degenerate baselines reduce parallax and destabilise triangulation [10]; and textureless surfaces—common on human skin, clothing, and manipulation objects—yield too few reliable correspondences for bundle adjustment to converge [14, 7]. The result is either outright reconstruction failure or incomplete, drift-prone geometry that cannot support downstream tasks such as grasping or object tracking.

Learning-based spatial models such as DUSt3R [14], PI3, and PI3X [15] overcome many of these failure modes by directly regressing dense 3D pointmaps from images, using learned priors to fill in low-texture and low-overlap regions where classical methods collapse. These models achieve strong per-frame accuracy—PI3 and PI3X in particular deliver the highest accuracy and completeness at a strict 1 cm robotics threshold among all models we evaluate. As purely spatial estimators, however, they process each frame independently and provide no explicit mechanism for enforcing temporal consistency across a sequence. The resulting frame-to-frame instability—manifesting as high-frequency jitter, scale fluctuations, or pose inconsistencies—can render otherwise accurate reconstructions unsuitable for downstream tasks including trajectory estimation, object tracking, and robotic manipulation.

Geometry-refinement approaches such as GGPT [2] take a different direction: rather than addressing temporal stability, they seek to improve the multi-view geometric consistency of per-frame feed-forward reconstructions by grounding dense pointmap predictions with sparse SfM-derived geometric guidance. Under the sparse-view dynamic conditions considered in this thesis, however, GGPT worsens Chamfer distance by 3–10× across all evaluated backbone models, making it unsuitable for geometry-critical applications in this setting. Dedicated 4D architectures—VGGT4D [6] and MonoFusion [16]—enforce sequence-level consistency, but at a measurable cost:

VGGT4D inflates Chamfer distance by 20–160 % relative to its per-frame baseline, and MonoFusion suffers systematic alignment convergence failures in sparse-view settings, producing duplicated subject artefacts rather than coherent reconstructions.

None of these approaches simultaneously achieves geometric fidelity and temporal consistency under sparse-view dynamic conditions. These results motivate the central design principle of this work: temporal consistency and geometric fidelity are not in fundamental conflict—they simply require a targeted rather than global smoothing strategy.

1.2 Problem Statement and Contributions

We address the problem of temporally consistent, geometrically precise 4D reconstruction from sparse multi-view video (two to four cameras) of dynamic scenes, evaluated on the DEX-YCB [1] and Hi4D [17] datasets. The main contributions of this work are:

- **Unified 4D Benchmarking Framework.** A structured evaluation pipeline that subjects six base models—MASt3R-SfM, VGGT, PI3, PI3X, VGGT4D, and MonoFusion—to identical data-loading, semantic masking, and metric-computation conditions across DEX-YCB [1] and Hi4D [17]. The framework measures both 3D per-frame and 4D sequence-level performance via Chamfer distance, accuracy and completeness at a 1 cm threshold, temporal jitter, drift, and camera pose error, with static/dynamic decomposition where applicable (see Chapter 3). It additionally evaluates the effect of GGPT [2] geometry refinement applied on top of all base models except MonoFusion.

- **Comparative Temporal Alignment Study.** An analysis of three strategies for lifting per-frame 3D reconstructions into a globally consistent 4D representation (see Chapter 3). The Reference strategy aligns every frame to a common anchor using static-scene correspondences and the Umeyama similarity transform. The Hierarchical strategy builds a binary tree of pairwise alignments, propagating consistency bottom-up. The Pose-Graph Optimisation strategy computes all $\mathcal{O}(T^2)$ pairwise edges, weights them by alignment reliability, and refines all frame poses simultaneously via robust geodesic averaging on SO(3). The Hierarchical strategy consistently delivers the best static-scene completeness and geometric stability; Pose-Graph Optimisation minimises the temporal consistency cost $\Delta_{\text{consistency}}$ (Equation 1.1) and best preserves dynamic-region accuracy.

$$\Delta_{\text{consistency}} = \text{Chamfer}_{\text{4D}} - \text{Chamfer}_{\text{3D}} \tag{1.1}$$

This quantity isolates the spatial degradation introduced purely by coordinate alignment, enabling direct comparison of alignment strategies and model families.

- **Confidence-Weighted Temporal Smoothing (Opt).** A ground-truth-free post-processing method that reduces frame-to-frame instability for any per-frame estimator without retraining (see **??**). Opt applies a Gaussian temporal filter to already-aligned pointmaps, weighted by model confidence and restricted to pixels classified as static by SAM2 [9]-derived masks—leaving dynamic pixels untouched to avoid suppressing real motion. A Pareto-frontier sweep over temporal bandwidth $\sigma \in \{0.0, 0.5, 1, 2, 4, 6, 8, 12, 20, 30\}$ and blending strength $\alpha \in \{0.0, 0.2, 0.5, 0.8, 1.0\}$ demonstrates that the optimal configuration is model-dependent: MASt3R-SfM benefits from a wider temporal window (σ=6, α=1.0), PI3 and PI3X from a moderate window (σ=4, α=1.0), and VGGT from a narrower window (σ=2, α=1.0), reflecting distinct jitter profiles across architectures. These per-model configurations achieve a 25–81 % jitter reduction while preserving geometry to within ±2–8 % Chamfer distance—a strictly superior trade-off to every native 4D architecture and geometry-refinement baseline evaluated in this thesis.

- **Empirical Characterisation of Architectural Families.** A systematic comparison of spatial-first models (PI3, PI3X, MASt3R-SfM, VGGT), the geometry-refinement approach GGPT, and native 4D architectures (VGGT4D, MonoFusion) reveals distinct and quantifiable failure modes under sparse-view dynamic conditions. PI3 and PI3X consistently achieve the strongest geometric performance across both DEX-YCB and Hi4D. MASt3R-SfM exhibits the greatest frame-to-frame instability, making it the model that benefits most from temporal refinement. VGGT provides a competitive balance between accuracy and stability. GGPT worsens Chamfer distance by 3–10× and produces inconsistent accuracy and completeness behaviour across architectures and view counts, while suppressing jitter to near-zero—a combination that makes it unsuitable for geometry-critical applications despite its apparent temporal stability. Native 4D architectures reduce temporal noise but consistently sacrifice geometric accuracy relative to spatial-first models. Together these findings establish the decoupled framework—strong spatial estimation, robust temporal alignment, and confidence-weighted temporal smoothing—as the most effective approach for sparse-view 4D reconstruction.

2

Literature Review and Background

This chapter reviews the theoretical and technical foundations underlying the reconstruction pipelines evaluated throughout this thesis. Section 2.1 introduces classical Structure-from-Motion (SfM) and Multi-View Stereo (MVS) methods, highlighting both their strengths and their limitations under sparse-view conditions. Section 2.2 surveys recent foundation models for 3D vision, grouped according to their treatment of spatial and temporal information. Section **??** describes the Segment Anything Model 2 (SAM2), which is used throughout this work to distinguish static and dynamic scene regions. Finally, Section **??** presents the mathematical concepts required for the proposed alignment and temporal smoothing framework, including similarity transformations, rotation averaging on SO(3), robust geodesic averaging, and geometric evaluation metrics.

2.1 Structure-from-Motion and Multi-View Stereo Foundations

Three-dimensional scene reconstruction from multiple images has traditionally been addressed through the complementary paradigms of Structure-from-Motion (SfM) and Multi-View Stereo (MVS). Together, these methods form the foundation of modern photogrammetry systems and remain the dominant approach for producing accurate metric-scale reconstructions from calibrated or partially calibrated image collections [5, 10, 11].

SfM seeks to jointly estimate camera poses and scene structure from image observations. Given a collection of overlapping images, distinctive visual features are first detected and matched across views. Geometric verification is then used to reject inconsistent correspondences, after which camera poses and sparse three-dimensional points are estimated through triangulation and refined using bundle adjustment [5, 10]. Modern SfM systems such as COLMAP [10] achieve remarkable accuracy and robustness on large-scale reconstruction tasks, often producing camera trajectories with sub-pixel reprojection error.

While SfM recovers only a sparse representation of scene geometry, MVS extends this sparse reconstruction into a dense surface estimate. Given camera poses obtained from SfM, MVS algorithms estimate depth for individual pixels by enforcing photometric consistency across multiple views. These depth estimates are subsequently fused into dense point clouds or surface representations [11, 12]. Modern MVS systems employ sophisticated view-selection strategies and regularisation techniques that enable high-quality reconstructions across a wide range of scenes [11].

Despite their success, classical SfM and MVS pipelines rely fundamentally on the existence of reliable multi-view correspondences. Performance degrades substantially when these assumptions are violated. Sparse-view reconstruction represents a particularly challenging regime because the number of observations available for establishing geometric constraints is severely limited. When only two to four cameras are available, several failure modes become prominent.

First, low inter-camera overlap reduces the number of features that are simultaneously visible across views, limiting the quantity and quality of geometric correspondences available for pose estimation [5, 8]. Second, short or degenerate baselines reduce triangulation accuracy because depth uncertainty grows rapidly as the triangulation angle decreases [5, 10]. Third, textureless surfaces such as human skin, clothing, walls, and many manipulation objects provide few distinctive features for matching, causing correspondence estimation and bundle adjustment to become unstable [14, 7]. Dynamic scene content introduces an additional challenge because moving objects violate the static-world assumptions underlying both feature matching and bundle adjustment [3].

These limitations are particularly acute in robotics and human-centred perception scenarios, where sparse multi-camera systems are often preferred for reasons of cost, deployment complexity, and occlusion constraints, yet such configurations provide exactly the conditions under which traditional SfM and MVS become least reliable. The resulting reconstructions may be incomplete, geometrically inconsistent, or entirely unsuccessful, limiting their usefulness for downstream tasks such as object tracking, trajectory estimation, manipulation planning, and scene understanding.

2.2 Foundation Models for 3D and 4D Reconstruction

Rather than relying on geometric matching alone, modern foundation models learn strong scene priors from large-scale image collections, enabling them to recover plausible geometry even in regions with limited texture, weak overlap, or ambiguous correspondence structure [14, 7, 13]. Although these methods share a common goal of reconstructing scene structure from images, they differ substantially in how geometry and time are represented.

The models evaluated in this thesis are grouped into three architectural families according to their primary design objective. *Spatial reconstruction models* estimate scene geometry from a collection of images without explicit temporal modelling (MASt3R-SfM, VGGT, PI3, and PI3X). *Geometry refinement models* improve existing reconstructions through additional geometric reasoning (GGPT). *Temporal reconstruction models* jointly model geometry and motion to produce temporally coherent reconstructions (VGGT4D and MonoFusion). Together, these families represent the principal design directions currently being explored in modern 3D and 4D vision, and their differing assumptions about geometry, correspondence, and temporal consistency form the basis for the comparative evaluation presented in Chapters 3 and 5.

2.2.1 Spatial Reconstruction Models

Spatial reconstruction models focus on recovering accurate scene geometry from a set of observations without explicitly modelling temporal evolution. Their primary objective is geometric reconstruction, with camera poses, depth, or pointmaps inferred directly from one or more images using learned geometric priors. While these models can be applied independently to each frame of a video sequence, they provide no mechanism for enforcing temporal consistency across successive reconstructions.

MASt3R-SfM. MASt3R-SfM is a hybrid reconstruction pipeline that combines learned geometric reasoning with classical SfM optimisation [4]. The approach builds upon DUSt3R, which introduced dense pointmap regression as an alternative to traditional depth estimation and triangulation [14], and MASt3R, which further improved robustness by grounding image matching directly in predicted three-dimensional geometry [7]. MASt3R-SfM integrates these learned representations into a complete SfM pipeline, using geometry-aware correspondences to initialise camera poses and scene structure before applying global optimisation. By combining learned matching with bundle adjustment, the system achieves state-of-the-art performance on challenging unconstrained reconstruction benchmarks while retaining the geometric consistency advantages of classical SfM.

VGGT. VGGT (Visual Geometry Grounded Transformer) represents a further step toward fully end-to-end geometric reasoning [13]. Rather than predicting isolated pairwise reconstructions, VGGT processes multiple views jointly using a transformer architecture trained to recover cameras, depth, and scene structure within a unified framework. By learning global geometric relationships directly from large-scale data, VGGT reduces dependence on handcrafted reconstruction stages and achieves strong performance across a wide variety of reconstruction tasks. Its architecture reflects a broader trend toward general-purpose geometric foundation models capable of solving multiple three-dimensional vision problems within a single learned representation.

PI3 and PI3X. PI3 introduces a reference-free formulation of visual geometry reconstruction that removes the need for a designated anchor image [15]. Using a fully permutation-equivariant architecture, the model jointly predicts camera poses and dense point maps from an unordered set of images, making reconstruction inherently robust to input ordering and reference-view selection. By learning geometric priors from large-scale data, PI3 can recover accurate scene structure even in sparse-view settings where traditional correspondence-based methods become unreliable. PI3X extends this framework with improved point-map decoding, more reliable confidence estimation, optional conditioning on camera and depth priors, and support for approximate metric-scale reconstruction. These modifications improve reconstruction quality while preserving the reference-free geometric reasoning of the original model.

2.2.2 Geometry Refinement Models

A complementary class of methods seeks to improve reconstruction quality without replacing the underlying reconstruction backbone. Instead of generating geometry directly, these approaches operate as post-processing stages that refine an existing geometric prediction using additional evidence.

GGPT. GGPT (Geometry-Grounded Point Transformer) is a geometry-refinement framework designed to enhance feed-forward three-dimensional reconstructions through geometry-guided correction [2]. The method combines dense pointmap predictions with sparse geometric cues derived from an auxiliary SfM process, using a transformer-based refinement network to improve consistency and reconstruction quality. Conceptually, GGPT occupies a middle ground between classical optimisation-based reconstruction and purely feed-forward prediction: geometric structure is first estimated by a backbone model and subsequently refined using additional geometric evidence.

Unlike spatial reconstruction and temporal reconstruction architectures, GGPT does not define a complete reconstruction pipeline in its own right. Instead, it functions as a model-agnostic refinement stage that can be applied to the outputs of existing reconstruction systems. This design makes it particularly relevant for understanding whether reconstruction quality is best improved through stronger backbone architectures, explicit temporal modelling, or post-hoc geometric refinement.

2.2.3 Temporal Reconstruction Models

A common limitation of purely spatial reconstruction models is the absence of explicit temporal reasoning: each frame or image set is reconstructed independently, meaning that geometric predictions may vary between consecutive frames even when the underlying scene remains static. Temporal reconstruction models address this by incorporating motion information and sequence-level reasoning during inference, jointly modelling geometry and temporal evolution rather than treating each frame as an isolated input.

VGGT4D. VGGT4D extends the VGGT architecture to four-dimensional reconstruction by incorporating temporal motion cues into the transformer backbone [6]. The model is designed to reason jointly about spatial structure and temporal evolution, enabling reconstruction of dynamic scenes across time. By leveraging information from neighbouring frames, VGGT4D aims to improve temporal consistency and reduce frame-to-frame fluctuations that commonly arise in per-frame reconstruction systems.

MonoFusion. MonoFusion approaches sparse-view four-dimensional reconstruction from a different perspective [16]. Instead of relying solely on multi-view geometric constraints, it combines monocular depth priors with temporal fusion mechanisms to accumulate information across a sequence. This strategy enables reconstruction from extremely sparse camera configurations and provides an explicit mechanism for enforcing temporal coherence. The model reflects a growing trend toward integrating learned monocular geometry with multi-view reasoning to overcome the limitations of sparse observations.

While temporal architectures address an important limitation of spatial models by introducing sequence-level consistency, the incorporation of temporal information inevitably introduces additional modelling assumptions and complexity. This creates a trade-off between geometric precision and temporal stability that remains an active area of research, and one that the comparative evaluation in this thesis aims to illuminate under controlled sparse-view conditions.

3

Methodology: The Evaluation Framework

4
System Implementation

5 Results

6
Refinement Strategy

7
Discussion

8

Conclusions

Bibliography

[1] Yu-Wei Chao *et al.* “DexYCB: A Benchmark for Capturing Hand Grasping of Objects”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021.

[2] Yutong Chen *et al.* “GGPT: Geometry-Grounded Point Transformer”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.

[3] Andrea Porfiri Dal Cin *et al.* “AnyMap: Learning a General Camera Model for Structure-from-Motion with Unknown Distortion in Dynamic Scenes”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2025, pp. 16674–16684.

[4] Bardienus Pieter Duisterhof *et al.* “MASt3R-SfM: a Fully-Integrated Solution for Unconstrained Structure-from-Motion”. In: *International Conference on 3D Vision 2025*. 2025. URL: https://openreview.net/forum?id=5uw1GRBFoT.

[5] R. I. Hartley and A. Zisserman. *Multiple View Geometry in Computer Vision*. Second. Cambridge University Press, ISBN: 0521540518, 2004.

[6] Yu Hu *et al.* *VGGT4D: Mining Motion Cues in Visual Geometry Transformers for 4D Scene Reconstruction*. 2025. arXiv: 2511.19971 [cs.CV]. URL: https://arxiv.org/abs/2511.19971.

[7] Vincent Leroy, Yohann Cabon, and Jerome Revaud. *Grounding Image Matching in 3D with MASt3R*. 2024. arXiv: 2406.09756 [cs.CV].

[8] Zador Pataki *et al.* *MP-SfM: Monocular Surface Priors for Robust Structure-from-Motion*. 2025. arXiv: 2504.20040 [cs.CV]. URL: https://arxiv.org/abs/2504.20040.

[9] Nikhila Ravi *et al.* “SAM 2: Segment Anything in Images and Videos”. In: *arXiv preprint arXiv:2408.00714* (2024). URL: https://arxiv.org/abs/2408.00714.

[10] Johannes Lutz Schönberger and Jan-Michael Frahm. "Structure-from-Motion Revisited". In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016.

[11] Johannes Lutz Schönberger *et al.* "Pixelwise View Selection for Unstructured Multi-View Stereo". In: *European Conference on Computer Vision (ECCV)*. 2016.

[12] Steven M. Seitz *et al.* "A Comparison and Evaluation of Multi-View Stereo Reconstruction Algorithms". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2006).

[13] Jianyuan Wang *et al. VGGT: Visual Geometry Grounded Transformer*. 2025. arXiv: 2503.11651 [cs.CV]. URL: https://arxiv.org/abs/2503.11651.

[14] Shuzhe Wang *et al.* "DUSt3R: Geometric 3D Vision Made Easy". In: *CVPR*. 2024.

[15] Zihan Wang *et al.* "MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 8252–8263.

[16] Zihan Wang *et al.* "MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 8252–8263.

[17] Yifei Yin *et al.* "Hi4D: 4D Instance Segmentation of Close Human Interaction". In: *Computer Vision and Pattern Recognition (CVPR)*. 2023.

List of Acronyms

A

Artifacts Availability

The artifacts associated with this thesis can be found in the ifi-GitLab[1]. Within this repository, there are two main folders: one containing the code to fine-tune the model, and the other containing the code to generate the training data, including all the different datasets. The README file provided in the repository contains step-by-step instructions on how to set up the environment used in this thesis, install all necessary dependencies, and run the source code.

Upload your Code to GitLab rather than attaching it to the PDF.

[1] https://gitlab.ifi.uzh.ch/ddis/Students/Theses/<NAME>

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